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ExportCo Network Security Project for
Coding Temple

PROJECT OVERVIEW

ExportCo is a small export company with 50 employees in HR, Marketing, and Finance.

The company's flat network exposed sensitive information to internal and external risks.

This project improves network security by implementing **VLAN segmentation**, **ACL-based access control**, and **secure remote management via SSH**.





Project Objectives

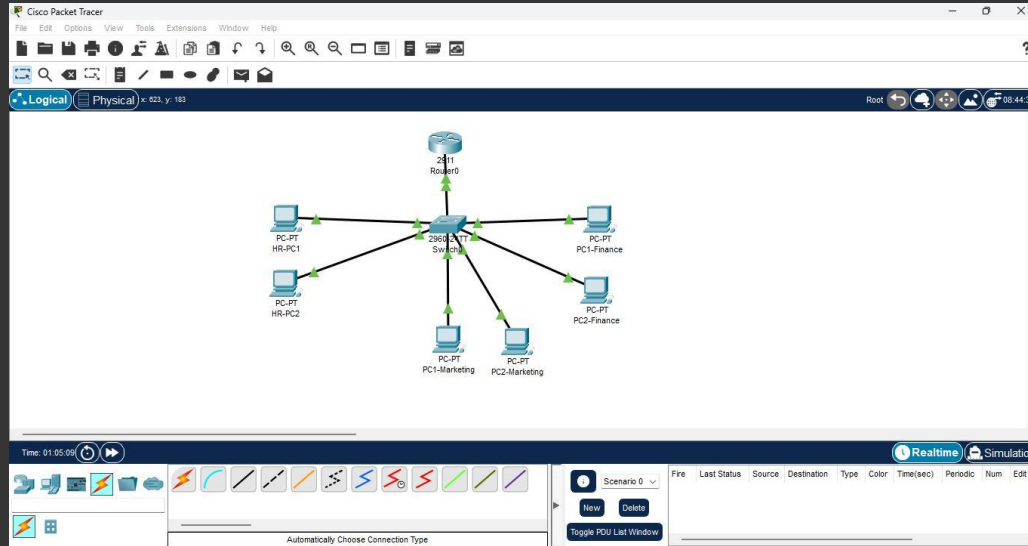
- Design VLANs for HR, Marketing, and Finance..
- Implement router-on-a-stick configuration for inter-VLAN routing
- Apply Access Control Lists (ACLs) to restrict unauthorized communication
- To enable secure remote management through SSH instead of Telnet.
- Test and validate the configuration.
- Document all steps clearly.
- provide recommendations for future improvements like MFA and IDS/IPS

Project Timeline

- ➔ **Phase 1 – Requirements Gathering (3 days)**
Identified risks, assets, and sensitive data.
- ➔ **Phase 2 – Network Design (4 days)**
Designed the secure network with VLAN segmentation and inter-VLAN routing.
- ➔ **Phase 3 – Implementation in Cisco Packet Tracer (6 days)**
Configured VLANs on the switch, router-on-a-stick, ACLs, and SSH.
- ➔ **Phase 4 – Testing and Validation (4 days)**
Ping tests, ACL validation, SSH verification, and Simulation Mode packet analysis.
- ➔ **Phase 5 – Documentation & Presentation (3 days)**
Wrote the final report, prepared screenshots, and built the presentation.



NETWORK TOPOLOGY



Here you can see the network design. I created three VLANs, one for each department: HR, Marketing, and Finance.

Each VLAN uses a different subnet for better isolation.

The router uses sub-interfaces to route traffic between VLANs using the router-on-a-stick method. ACLs control which departments can communicate.

ExportCo Risk

ID	Risk / Threat	Likelihood	Impact	Mitigation / Status
R1	Flat network allowed all departments (HR / Marketing / Finance) to access sensitive Finance data.	High	Critical	Segmented network into VLAN 10 / VLAN 20 / VLAN 30 with unique subnets and gateways. Status: Mitigated.
R2	Marketing could reach Finance resources and view financial information.	Medium	High	
R3	Telnet / cleartext admin access could leak router credentials.	High	Critical	Telnet disabled. SSH v2 enforced with RSA keys and privileged admin account. Status: Partially Mitigated
R4	Limited perimeter security and monitoring increases external attack / lateral movement risk	Medium	High	
R5	No centralized logging / DLP means possible data exfiltration of Finance or customer data	Medium	High	Recommend SIEM logging. DLP, secure backups, and Security Awareness Training. Status: Open
R6	Users without security training could fall for phishing or misuse credentials	Medium	Medium	

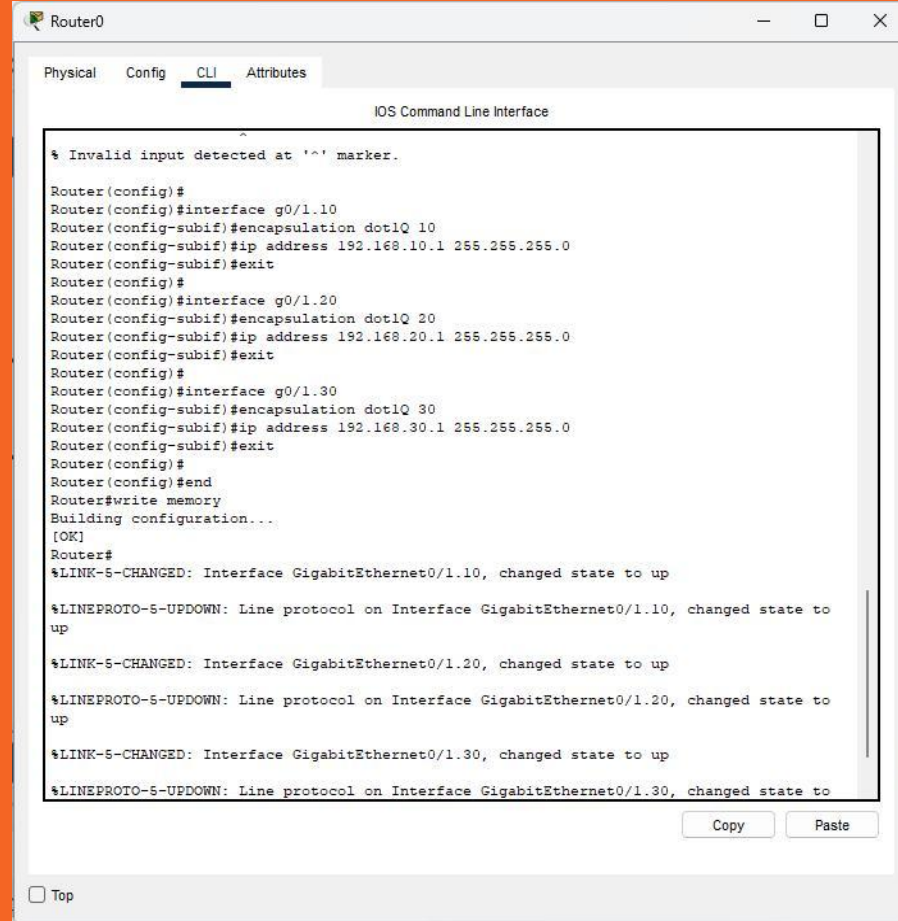
VLAN and Router Configuration

VLANs were created for each department to keep the network organized and secure.

The router uses subinterfaces to connect the VLANs and control traffic between them.

This setup allows each team (HR, Marketing, Finance) to work safely on their own network while staying connected to the company's system.

This improve Security and Performance (reduce internal threats and makes the network easy to manage)



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

% Invalid input detected at '^' marker.

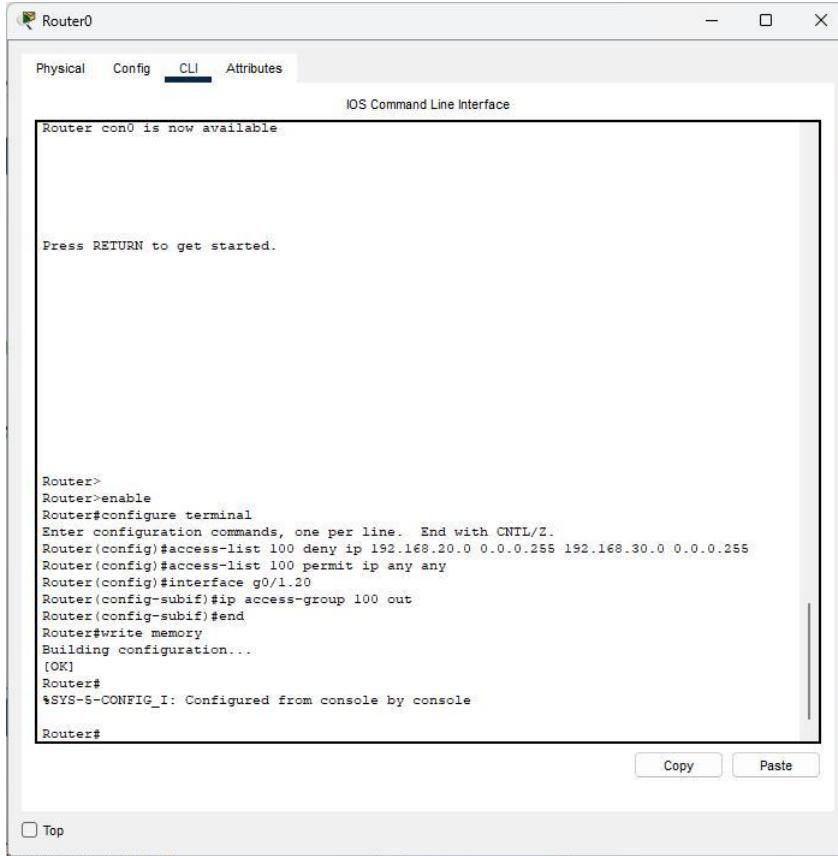
Router(config)#
Router(config)#interface g0/1.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface g0/1.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#interface g0/1.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#end
Router#write memory
Building configuration...
[OK]
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up

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```

Access Control List (ACLs)



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router con0 is now available

Press RETURN to get started.

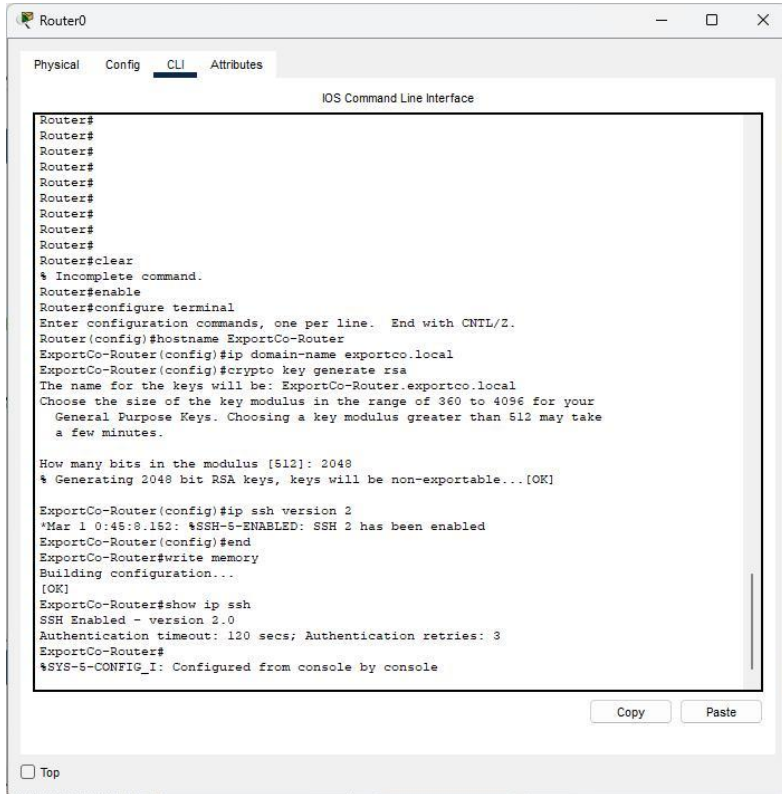
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/1.20
Router(config-subif)#ip access-group 100 out
Router(config-subif)#end
Router#write memory
Building configuration...
[OK]
Router#
$SYS-5-CONFIG_I: Configured from console by console
Router#
```

Access Control Lists (ACLs) were used to control communication between departments.

The ACL denies traffic from the Marketing VLAN (192.168.20.0/24) to the Finance VLAN (192.168.30.0/24), while allowing all other connections.

This setup enforces the principle of least privilege, ensuring that only authorized traffic moves through the network and preventing internal data leaks (Users will have access only on what they need)

SSH – Secure Remote Access



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#clear
% Incomplete command.
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ExportCo-Router
ExportCo-Router(config)#ip domain-name exportco.local
ExportCo-Router(config)#crypto key generate rsa
The name for the keys will be: ExportCo-Router.exportco.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

ExportCo-Router(config)#ip ssh version 2
*Mar 1 0:45:0.152: %SSH-5-ENABLED: SSH 2 has been enabled
ExportCo-Router(config)#end
ExportCo-Router#write memory
Building configuration...
[OK]
ExportCo-Router#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
ExportCo-Router#
%SYS-5-CONFIG_I: Configured from console by console
```

SSH (Secure Shell) was configured on the router to allow encrypted remote management.

A local admin user and RSA keys were created for secure login.

Telnet was disabled to prevent unencrypted access.

This ensures that only authorized administrators can manage the network safely.

Testing Connectivity

Network connectivity was tested using ping commands between departments.

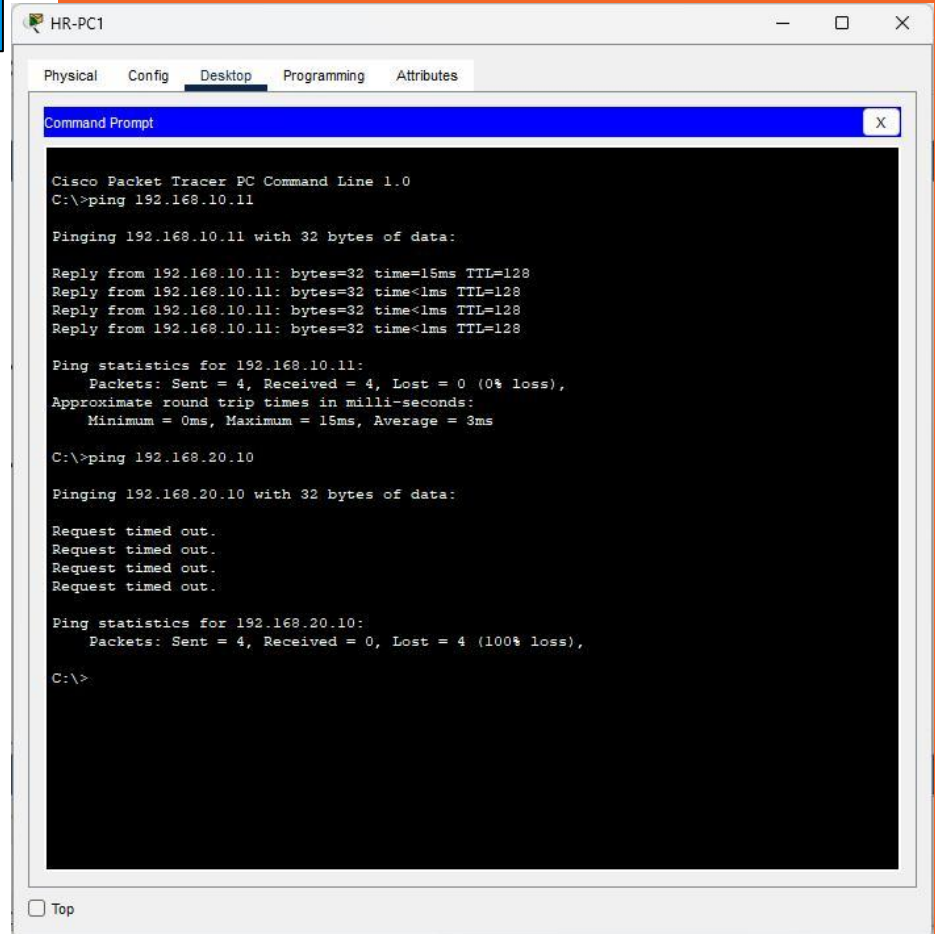
Devices inside the same VLAN could communicate successfully.

Pings between different VLANs were blocked by ACLs, confirming that access control works correctly.

This test proved that network segmentation and ACL rules were implemented successfully.

✓ HR → HR = Success

✗ HR → Finance = Blocked



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named HR-PC1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=15ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

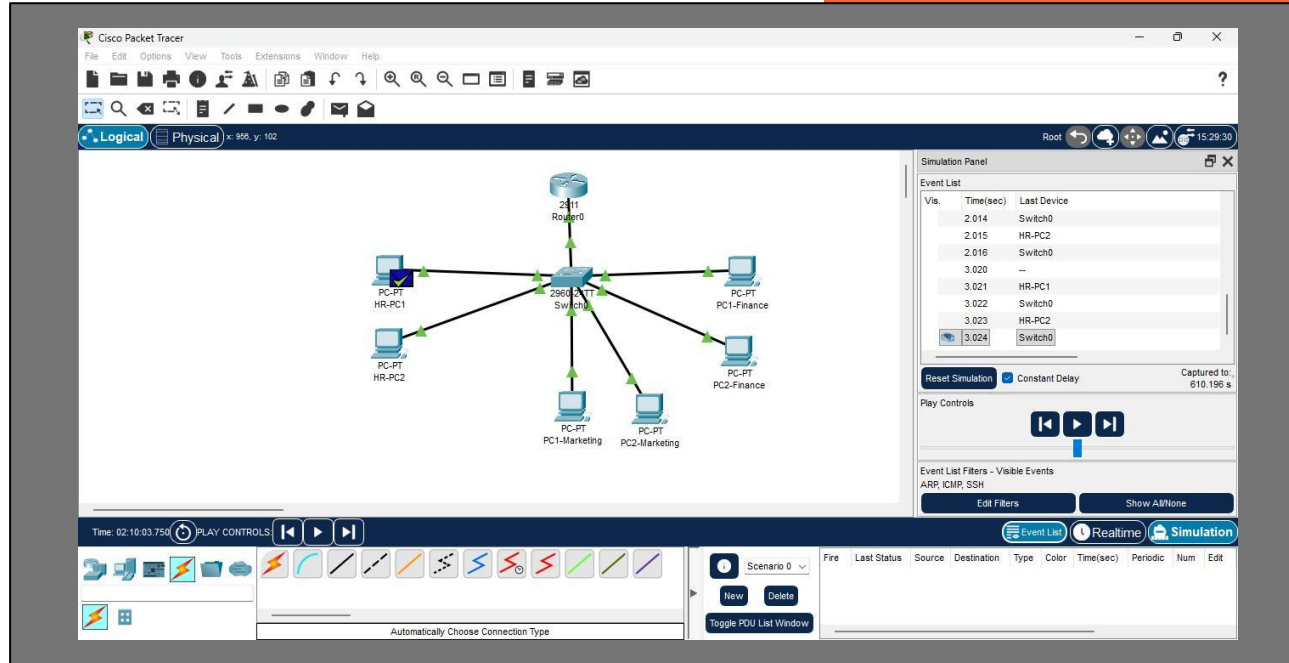
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

At the bottom of the window, there is a checkbox labeled "Top" which is currently unchecked.

Wireshark Simulation Analysis



Filters Used

 **ICMP (Ping)** → Tested connectivity between VLAN devices.

 **ARP (Address Resolution Protocol)** → Verified MAC and IP mapping during communication.

 **SSH (Secure Shell)** → Confirmed encrypted remote connections to the router.

Using Cisco Packet Tracer's Simulation Mode, different types of traffic were analyzed to validate communication between VLANs.

The ICMP packets show successful communication inside the HR VLAN, while ARP packets confirm proper MAC and IP mapping.

This test simulated Wireshark behavior and verified that the ACLs correctly block inter-VLAN traffic while allowing internal communication.



Results and Analysis

- ✓ The network was successfully segmented into secure VLANs.
- ✓ ACLs blocked unauthorized inter-VLAN traffic.
- ✓ SSH ensured encrypted administrative access.
- ✓ Ping tests confirmed proper Layer 3 communication.

BEFORE

Flat, unprotected network

No access control

Unsecured telnet

No monitoring

AFTER

Segmented VLANs

ACLs applied

Secure SSH

Wireshark validation

ExportCo Incident Response Playbook

High-level process to detect, contain, and recover from a security incident

ID	Phase / Step	Goal	Key Actions	Status
IR1	Detection / Identification	Identify a real security incident ASAP	• SIEM alert or employee reports suspicious activity • Validate if the alert is real or a false positive • Identify affected user / device / system	In Progress
IR2	Containment	Stop the attack from spreading	• Isolate affected device (move to Guest VLAN / block switch port) • Disable compromised accounts / reset exposed credentials • Limit lateral movement using VLAN segmentation and ACLs	In Place (network segmentation + ACLs)
IR3	Eradication	Remove the root cause	• Remove malware / malicious tools • Patch vulnerabilities • Disable insecure services (Telnet off → SSH only with RSA keys) • Force password reset for leaked credentials	Partially In Place
IR4	Recovery	Safely restore normal operations	Reconnect ONLY if device is compliant (firewall on, encryption / BitLocker enabled, antivirus active, Intune policy Compliant and Monitor logs and network traffic for abnormal behavior)	In Progress

Goal: Protect Finance data, reduce downtime, and improve future response.

RECOMMENDATIONS AND LESSONS I LEARNED



In this final stage, I summarized the key lessons learned during the implementation process. VLANs, ACLs, and SSH were essential to strengthening network segmentation and secure access.

Additionally, I proposed next steps such as adding MFA, IDS/IPS systems, and employee awareness training to further improve ExportCo's security posture.

This project taught me the importance of detailed documentation and continuous monitoring to maintain a secure and efficient infrastructure.

✓ This project demonstrated the importance of secure network design using VLANs, ACLs, and SSH.

✓ All configurations improved segmentation, control, and secure management.

✓ Wireshark testing validated successful communication and security.

Keep the world secure. Stay cyber smart.

💡 Thank you for your attention!

